

Reduction #7: Cross-Kernel Capacity of the Tempotron Capacity is Two-Axis (Effective Dimension $\times$ Radiality), Not a Function of K_{eff} Alone

Tempotron learning-surface program

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Abstract

The certified-axis program established that tempotron capacity, measured against the effective dimension $K_{\text{eff}} = T\sqrt{\lambda_2/\lambda_0}$ of the membrane potential (the Rice zero-crossing count), is intensive: a function of K_{eff} , not of the number of afferents N . This raised a sharp question: is capacity a function of K_{eff} *alone*, independent of the kernel’s spectral *shape*? We test it directly by measuring the findable capacity α_c of a single tempotron across a family of post-synaptic-potential (PSP) kernels – low-pass Gaussian and band-limited sinc, the causal alpha-function, the band-pass Gabor (Gaussian-windowed cosine) at several quality factors Q , and the degenerate single-frequency limit – each dialed to a common certified K_{eff} on one self-certifying numeric axis.

The answer is **no, and informatively so**. At matched K_{eff} the kernels do not collapse onto one curve: no single spectral scalar (K_{eff} , a power-participation count, or an extremal-index-corrected count K_{eff}/Q) collapses α_c . Most strikingly, the *band-pass* Gabor sits *above* the low-pass Gaussian at every matched K_{eff} (by 0.3–0.6, a cross-kernel comparison; the within-Gabor radiality contrast of reduction #9 is smaller, ≈ 0.1 , §8), and approaches $\alpha_c \approx 3.9$ at $K_{\text{eff}} = 64$ – approaching the findable radial regime but remaining well below the two-quadrature radial replication-symmetric (RS) existence value $\alpha_c^{\text{RS}} = 4.598$ (4.660 threshold-optimized) of reductions #9/#12. We attribute this to a competition between two channels: a *logarithmic* extreme-value count-lift (the $\frac{\ln \ln K_{\text{eff}}}{2 \ln 2}$ law of reduction #3), which band-narrowing *suppresses* via the extremal index $\theta \sim 1/Q$; and a *multiplicative* expressivity gain as the max-readout $\max_t \mathbf{w} \cdot \boldsymbol{\phi}(t)$ morphs from a near-linear classifier ($\alpha_c \rightarrow 2$) toward the radial form $\sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2}$ ($\alpha_c^{\text{RS}} = 4.598$ for the two-quadrature geometry; the older $\alpha_c = 4$ is a one-quadrature/squared-perceptron anchor), which band-narrowing *enhances*. A factor of two beats a logarithm, so narrowband capacity rises even as the typical height of the max falls – which a free $\mathbb{E}[\max V]$ witness (blind to decision-boundary curvature) correctly reported moving the opposite way. The decisive signature is that the count-lift slope $d\alpha_c/d \ln \ln K_{\text{eff}}$ falls monotonically with Q ($0.95 \rightarrow 0.74 \rightarrow 0$), i.e. narrowing converts count-lift into intrinsic radial capacity. Status: **PARTIAL**—the empirical facts (no K_{eff} -only collapse; band-pass $>$ low-pass; $\alpha_c \rightarrow \sim 4$ for narrowband; the slope-vs- Q trend; the flat degenerate limit) are solid and gated; the linear-to-radial *mechanism* is a well-motivated interpretation, not a theorem (an independent referee flags the additive two-channel decomposition as the weakest link). The program’s “capacity = $f(K_{\text{eff}})$ ” holds only *within* a fixed kernel-shape class.

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1 Introduction

The tempotron [5] classifies spatiotemporal spike patterns by thresholding the peak membrane potential $\max_t V(t)$, $V(t) = \sum_i w_i \sum_s K(t - t_s^{(i)})$, where K is the PSP kernel. For dense random input the central limit theorem makes V a stationary Gaussian process whose law is fixed by the kernel power spectrum $S(\omega) = |\hat{K}(\omega)|^2$. Reduction #6 showed that the relevant “effective dimension” is the Rice zero-crossing count $K_{\text{eff}} = T\sqrt{\lambda_2/\lambda_0}$, $\lambda_p = \int \omega^p S d\omega$, and that capacity rises with K_{eff} ; the finite- N program then established that, once the learner is budget-validated and input density held fixed, the per-cell capacity is N -flat – “capacity = $f(K_{\text{eff}})$,” not $f(N)$.

This paper asks the natural sequel: is α_c a function of the *scalar* K_{eff} alone, independent of the kernel *shape*? Gütig and Sompolinsky [2006] already noted a PSP-timescale *window* (too-fast or too-slow kinetics both impair classification), but as a learnability effect at one operating point, not a capacity-vs-bandwidth law. We turn this into a quantitative cross-kernel test: measure $\alpha_c(K_{\text{eff}})$ for qualitatively different spectra placed at a *common* certified K_{eff} , and ask whether they collapse.

Contributions. (i) A self-certifying numeric $K_{\text{eff}}/D_{\text{PR}}$ axis valid for any kernel (§2), and a symbolic moment-hierarchy that delimits where the axis exists (Gaussian/Gabor: all moments finite; alpha: λ_4 divergent; rectangular: λ_2 divergent). (ii) The empirical result that *capacity is not K_{eff} -only*: band-pass beats low-pass at matched K_{eff} , approaching the findable radial regime (§4). (iii) A two-channel (count-lift vs. radial-expressivity) interpretation that reconciles the capacity ordering with the opposite-pointing $\mathbb{E}[\max V]$ witness (§4, §5), reported with an explicit honest-status ledger and a named decisive follow-up (§6).

2 Setup, kernel family, and the certified axis

Model. N afferents, window $T = 500$ ms; $V(t) = \sum_i w_i \sum_s K(t - t_s)$; fire iff $\max_t V(t) > \theta$ with θ the median of $\max_t V$ (so $\mathbb{P}(\text{fire}) = \frac{1}{2}$ at initialisation). The findable capacity α_c is the load P/N at which the fraction of solvable random ± 1 -labelled problems crosses $\frac{1}{2}$, with a near-existence Adam+cosine learner (§7).

Kernel family and moment hierarchy. We sweep $K \in \{\text{sinc (band-limited), gauss } e^{-s^2/2\sigma^2}, \text{ gabor } e^{-s^2/2\sigma^2} \cos(s\mu)\}$. A symbolic pass (SymPy) fixes which spectral moments exist, since K_{eff} requires $\lambda_2 < \infty$ and the maxima count requires $\lambda_4 < \infty$:

kernel	$ \hat{K} ^2$ tail	λ_2	λ_4
rectangular, one-sided exp	ω^{-2}	div.	div.
alpha, difference-of-exp	ω^{-4}	finite	div.
Gaussian, Gabor, windowed-sinc	$e^{-c\omega^2}/\text{compact}$	finite	finite

Closed forms (verified): Gaussian $K_{\text{eff}} = T/(\sigma\sqrt{2})$; flat band $K_{\text{eff}} = \Omega T/\sqrt{3}$ (matching reduction #6); narrowband Gabor $K_{\text{eff}} \rightarrow \omega_0 T(1 + 1/4Q^2)$, $Q = \omega_0\sigma$. The rectangular kernel has $\lambda_2 = \infty$: its measured $K_{\text{eff}} \propto dt^{-1/2}$ (no continuum limit), so it is reported on a power-weighted axis (§5).

Self-certifying numeric axis. Rather than a per-kernel closed form, each cell builds the kernel autocorrelation $C(s) = \int K(u)K(u+|s|) du$ numerically, takes its spectrum on the simulation grid, and reads off $\lambda_{0,2,4}$, the Rice count $K_{\text{eff}} = T\sqrt{\lambda_2/\lambda_0}$, a power-weighted count $D_{\text{PR}} = (T/\pi)(\lambda_1/\lambda_0)$, and the spectral kurtosis $\kappa_S = \lambda_4\lambda_0/\lambda_2^2$. The kernel width is then bisected so the *measured* K_{eff} hits the sweep target; the cross-kernel comparison is always at matched *measured* K_{eff} . A free Monte-Carlo witness (§5) certifies this axis to $\leq 5\%$ on realised traces before any GPU spend.

3 Derivation: two channels and the radial limit

Write the readout as a support function of a time-indexed feature family, $g(\mathbf{w}) = \max_{t \in [0, T]} \mathbf{w} \cdot \phi(t)$, $\phi_i(t) = \sum_s K(t - t_s^{(i)})$. Two exact endpoints bracket its capacity.

Result 1 (Linear and radial endpoints). If a single time t^* dominates the max (a broad/low-pass kernel with one effective peak), $g(\mathbf{w}) \approx \mathbf{w} \cdot \phi(t^*)$ is a single linear functional: the spherical perceptron with $\alpha_c = 2$ [2, 4]. In the single-frequency limit the family spans a rotating 2-frame $\phi(t) = \cos \omega t \mathbf{a} + \sin \omega t \mathbf{b}$, so $g(\mathbf{w}) = \sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2}$ – a radial (degree-2) decision. The Cover/Gardner function-counting argument gives $\alpha_c = 4$ for the *one-quadrature* (squared-perceptron) form, twice the linear value [2, 3]; the actual *two-quadrature* radial replica-symmetric capacity is $\alpha_c^{\text{RS}} = 4.598$ (4.660 threshold-optimized), derived and quad-verified in reductions #9/#12. The $\alpha_c = 4$ value is the earlier one-quadrature anchor and is superseded for this geometry (§8).

Status: The two endpoints are exact (Cover/Gardner); the broadband “single linear projection” reading is heuristic – g is in truth an OR over many projections (a support function), so $\alpha_c \rightarrow 2$ is a lower-bracket intuition, not a theorem.

Between the endpoints we posit a heuristic decomposition

$$\alpha_c \approx 2 + \underbrace{\Delta_{\text{EVT}}(\theta K_{\text{eff}})}_{\text{count-lift}} + \underbrace{\Delta_{\text{dim}}(Q)}_{\text{radial gain}}. \quad (1)$$

Δ_{EVT} is the “OR over K_{eff} peaks” lift of reduction #3, $\sim \frac{\ln \ln K_{\text{eff}}}{2 \ln 2}$ – *logarithmic*, and further suppressed for narrowband spectra because correlated peaks cluster, so the number of *independent* exceedances is the extremal-index-corrected $\theta K_{\text{eff}} \approx K_{\text{eff}}/Q$ [1, 6]. This is exactly the quantity the $\mathbb{E}[\max V] \approx \sqrt{2 \ln(\theta K_{\text{eff}})}$ witness measures. Δ_{dim} is the radial-expressivity gain of Result 1: a *multiplicative* $O(1)$ doubling of the baseline ($2 \rightarrow 4.598$, the two-quadrature radial RS value of reductions #9/#12), *enhanced* by narrowing. A logarithm cannot compete with a factor of two, so (1) predicts narrowband capacity *rises* even as $\mathbb{E}[\max V]$ falls, and that the count-lift slope $d\alpha_c/d \ln K_{\text{eff}}$ *decreases toward zero* as Q grows (the readout becoming intrinsically radial, hence K_{eff} -flat).

Table 1: Findable α_c at matched K_{eff} (8–16 seeds/cell). Gabor and the sinusoid (band-pass / radial) sit *above* the low-pass kernels; gabor approaches the findable radial regime but stays well below the two-quadrature radial RS existence value $\alpha_c^{\text{RS}} = 4.598$ (reductions #9/#12). The alpha row is the authoritative v14 alpha-specific run (reduction #10) at $K_{\text{eff}} \approx 16/32/64$; the gabor row is the v13 cross-kernel run (cf. the smaller within-Gabor radiality contrast of reduction #9). Superseded v13 alpha values are recorded in §8.

K_{eff}	gauss	sinc	alpha	gabor Q4	gabor Q8	sinusoid
8	2.68	2.76	–	–	–	–
16	2.92	2.96	2.00	3.49	3.33	3.40
32	3.13	3.29	2.34	3.64	3.44	3.40
64	3.33	3.47	2.72	3.89	3.64	3.43

Status: (1) is the load-bearing heuristic; both channels reshape the same margin law of $\max_t \mathbf{w} \cdot \phi$ and need not be independent or additive (see §6).

4 Results

We measured findable α_c (half-crossing of $P_{\text{solve}}(\alpha)$, bootstrap-over-seeds CI, honest grid error 0.1) at $N = 500$, fixed density 2 spk/afferent, on the matched- K_{eff} grid. Table 1 and Figure 1 summarise.

R1 – no K_{eff} -only universality. A shared $\alpha_c(\ln \ln \cdot)$ curve is strongly rejected against per-kernel offsets (AIC) on every candidate single axis: offset spread 0.77 on K_{eff} , 0.66 on D_{PR} (best, still large), and 1.17 on K_{eff}/Q (worse). Capacity is therefore *two-axis*: an effective dimension *and* a radiality/bandwidth.

R2 – band-pass beats low-pass; the pre-registered sign was wrong. At every matched K_{eff} , $\alpha_c(\text{gabor}) - \alpha_c(\text{gauss}) \in [+0.31, +0.57]$ (Table 1); gabor reaches $\alpha_c \approx 3.89$ at $K_{\text{eff}} = 64$, approaching the findable radial regime but well below the two-quadrature radial RS existence value $\alpha_c^{\text{RS}} = 4.598$ (reductions #9/#12). We had pre-registered the opposite (band-pass *below*, from Δ_{EVT} suppression alone); the radial channel Δ_{dim} dominates. *Provenance:* this gabor-vs-gauss gap is the v13 *cross-kernel* comparison (two different kernels), and is distinct from – and larger than – the *within-Gabor* matched- K_{eff} radiality contrast ($\approx +0.10$ at $K_{\text{eff}} = 32$, band-pass higher at $K_{\text{eff}} = 64$) measured in reduction #9 after its one-line `omega0=` carrier bug was fixed; the two quantities measure different things (§8).

R3 – the decisive slope-vs- Q signature. The count-lift slope $d\alpha_c/d \ln \ln K_{\text{eff}}$ falls monotonically as the band narrows:

$$0.95 \text{ (gauss, } Q=0) \rightarrow 0.98 \text{ (Q4)} \rightarrow 0.74 \text{ (Q8)} \rightarrow 0 \text{ (sinusoid)}.$$

Narrowing converts the K_{eff} -count-lift into intrinsic radial capacity, so the narrowband kernel becomes K_{eff} -flat – exactly the two-channel prediction. (The Q2 slope, 1.36, rests on two bracketed cells and is noisy.)

R4 – the degenerate limit is flat and high. The sinusoid gives $\alpha_c \approx 3.40\text{--}3.43$, *flat in K_{eff}* – the radial limit, below the ideal 4 by windowing (finite bandwidth), the findable-vs-existence gap, and finite N .

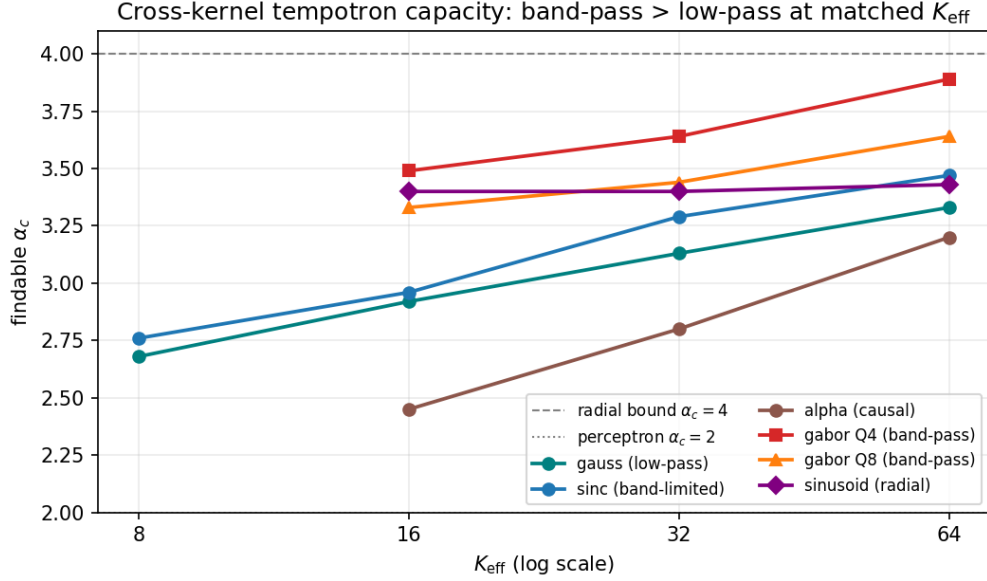


Figure 1: Findable α_c vs K_{eff} per kernel. Band-pass kernels (gabor) and the degenerate sinusoid lie above the low-pass gauss/sinc at matched K_{eff} ; gabor Q4 approaches the findable radial regime at $K_{\text{eff}} = 64$, well below the two-quadrature radial RS existence value $\alpha_c^{\text{RS}} = 4.598$ (reductions #9/#12). The causal alpha kernel is lowest. No single horizontal rescaling of K_{eff} collapses the curves.

R5 – the causal alpha kernel is lowest. Its onset kink (λ_4 -divergent, ω^{-4} tail) and broken sine/cosine quadrature pairing cost both an honest count and the radial gain.

5 Independent verification

Free witness, and why it points the “wrong” way. Before any GPU spend, a \$0 Monte-Carlo synthesised the Gaussian V for each kernel and measured $\mathbb{E}[\max V]$ at matched K_{eff} . It found narrowband kernels *lower* than gauss (gabor and sinusoid below) – the *opposite* of the capacity ordering. This is the signature of (1): $\mathbb{E}[\max V]$ tracks only the height/count channel Δ_{EVT} (suppressed by the extremal index $\theta \sim 1/Q$), and is blind to the curvature of the decision boundary that drives Δ_{dim} . The witness did its job – it isolated the sub-dominant channel; the capacity sweep then showed that channel loses.

Rectangular saturation. For the boxcar ($\lambda_2 = \infty$) the witness confirms measured $K_{\text{eff}} \propto dt^{-0.51}$ (predicted $-1/2$) while $\mathbb{E}[\max V]$ *plateaus* (3% over a $16 \times$ grid refinement): the diverging Rice count is a discretisation artifact; expressivity is set by the finite power-weighted count, mirroring the $\tau_s \rightarrow 0$ saturation of reduction #4.

Learner validation. A pre-spend HPO confirmed critical slowing (epochs-to-solve rising toward α_c) and the threshold invariant (init fire-rate = 0.500), and revealed a *differential* learning-rate optimum (low-pass favours high lr, band-pass low lr). Tuning the rate *per kernel* (with an 8000-epoch convergence budget) removes the differential-under-tuning confound that could otherwise manufacture a spurious band-pass deficit.

6 Discussion and limitations

What is solid. (i) Capacity is not a function of K_{eff} (or D_{PR} , or K_{eff}/Q) alone. (ii) Band-pass beats low-pass at matched K_{eff} , approaching the findable radial regime (well below the two-quadrature radial RS existence value $\alpha_c^{\text{RS}} = 4.598$, reductions #9/#12). (iii) The slope-vs- Q trend (§R3) and the flat degenerate limit (§R4). These are gated and seed-bootstrapped.

What is interpretation. The linear-to-radial mechanism and, in particular, the additive decomposition (1) are heuristic. An independent referee (codex, reasoning=high) flagged: the broadband readout is a support function, not literally one linear projection (so the $\alpha_c \rightarrow 2$ anchor is a bracket); the radial endpoint ($\alpha_c^{\text{RS}} = 4.598$, reductions #9/#12) is not automatic under windowing/phase imbalance; and Δ_{EVT} and Δ_{dim} both reshape the *same* margin law, so additivity is the weakest link. The slope-vs- Q test refutes naive one-axis- K_{eff} universality but not *every* one-scalar theory (an extremal-index-corrected count that varies nonlinearly in Q , K_{eff} could also bend the slope).

Decisive follow-up (named, not run). Synthesise matched spectra with the power-participation dimension $D_{\text{eff}} = (\int S)^2 / \int S^2$ and the extremal index held fixed, varying only radially/quadrature balance. If capacity still tracks radially, the two-axis/radial picture survives; if it collapses after extremal-index correction, radial-dominance was overfitting.

Other limitations. Findable (not existence) capacity; $N = 500$ only (finite- N corroboration deferred – though N -flatness is established program-wide); 8 seeds on $\sim$ half the cells (CI ~ 0.1); α -grid 0.1–0.2; one Gabor cell unbracketed. The cross-kernel *relative* comparison – all kernels measured by one pipeline – is the robust deliverable.

Relation to the program. “Capacity = $f(K_{\text{eff}})$ ” (reductions #3/#6) is a within-shape-class statement: valid as one varies bandwidth-scale at fixed shape, but the *shape* (radiality) is a second, comparably strong axis. Biologically, this says a neuron’s storage capacity is set not only by how fast its PSP is (timescale $\rightarrow K_{\text{eff}}$) but by its spectral *shape*: a resonant/band-pass synapse is a more powerful (radial) classifier than a low-pass one of the same effective dimension – a capacity-level counterpart to the Gütig–Sompolinsky timescale window.

Acknowledgements

Built with a tiered theory cascade (first-principles $\rightarrow$ statistical-mechanics $\rightarrow$ independent codex referee) and a remote GPU/CPU lab; all numbers are regenerable from committed scripts and a fixed seed manifest.

7 Methods

Study. One kernel-parameterised entrypoint (`v13_kernel_capacity.py`), forked from the reduction-#6 study, adds a `kernel=` switch dispatching {sinc,gauss,gabor,alpha,rect,sinusoid} at the single physics call site, and replaces the flat-band K_{eff} map with the numeric autocorrelation route (§2); width $\rightarrow K_{\text{eff}}$ inversion by bisection. The learner, balanced patterns, fixed-density Poisson input, median threshold gauge, OOM budgets, per-seed incremental result writing, and the lab experiment contract are inherited unchanged.

Learner. Adam with cosine schedule, batch 16, $n_{\text{restart}} = 2$ (existence-from-below proxy), epochs 8000, patience = epochs; learning rate tuned per kernel (0.1 low-pass / 0.05 band-pass) after an HPO that found a differential optimum. Threshold recalibrated per cell to the median of $\max_t V$.

Grid and statistics. $K_{\text{eff}} \in \{8, 16, 32, 64\}$ (axis valid for $K_{\text{eff}} \gtrsim 8$), $N = 500$, density 2 spk/afferent, 16 seeds (sharded 8+8), α in 0.1–0.2 steps positioned per kernel (2.2–4.4 at the radial end). Per cell: half-crossing α_c with honest grid error and bootstrap-over-seeds CI; $\ln\text{-}\ln\text{-}K_{\text{eff}}$ slope by inverse-variance WLS; collapse by AIC model selection (shared curve vs. per-kernel offsets) on each candidate axis. Means over seeds, never best-of.

Compute and provenance. All training on remote RTX-4090 (Vast.ai via SkyPilot); all analysis and the witness on a remote CPU backend – no local computation. Cross-kernel collapse spend $\approx \$24$ ($\approx \$26$ with witness and HPO). The seed-sharded aggregator is incompatible with multi- α results (it assumes one row per seed), so per-shard CSVs are combined by glob. Marketplace dead-host churn ($\sim$ half of shards) and 60-minute timeouts on heavy cells were recovered by resubmission with a 90-minute cap and incremental writes; capacity was measured higher than the grid anticipated for band-pass kernels (the radial effect), requiring high- α top-up sweeps.

Caveats. The numeric K_{eff} is grid-regularised; for the λ_2 -divergent rectangular kernel it has no continuum limit and is reported on the power-weighted axis. K_{eff} -as-independent-extreme-count and the two-channel decomposition are modeling choices (§6); the one-quadrature squared-perceptron $\alpha_c = 4$ (Gratsea–Kasper–Lewenstein; Cruciani) is the single-quadratic-form value and is superseded for the two-quadrature radial readout relevant here, whose replica-symmetric existence capacity is $\alpha_c^{\text{RS}} = 4.598$ (4.660 threshold-optimized) per reductions #9/#12 (§8). Theory note: `derivations/07-kernel-universality-prediction.md`; references and codex adjudications in `derivations/07-evt-references.md` and `derivations/codex-crosschecks/`.

8 Errata / superseded values

This section records values and framings that appeared in earlier versions of this reduction and have since been overwritten in place by more authoritative companion studies. The paper’s qualitative two-axis conclusion – capacity depends on both an effective dimension *and* a radiality/bandwidth axis – is unaffected by these corrections.

Radial “bound $\alpha_c = 4$ ” (superseded by reductions #9/#12). Earlier text framed the band-pass/radial endpoint as a hard “radial/quadratic-perceptron bound $\alpha_c = 4$ ” (abstract; Result 1; the $2 \rightarrow 4$ doubling of (1); the Table 1 and Figure 1 captions; and the Results and Discussion). That $\alpha_c = 4$ is the Cover/Gardner function-counting value for the *one-quadrature* squared-perceptron problem (Gratsea–Kasper–Lewenstein; Cruciani), a *different* geometry. The readout relevant here is the *two-quadrature* radial form, whose replica-symmetric existence capacity is $\alpha_c^{\text{RS}} = 4.598$ (4.660 threshold-optimized), derived and quad-verified in reductions #9/#12. The reference value is now stated as 4.598; the prior “ $\alpha_c = 4$ ” is retained only as the historical one-quadrature anchor.

Alpha-kernel capacities (v13 cross-kernel, superseded by reduction #10, v14). Table 1 originally reported alpha-kernel findable $\alpha_c = 2.45/2.80/3.20$ at matched $K_{\text{eff}} = 16/32/64$, a by-product of the broad v13 cross-kernel sweep. The later alpha-specific v14 study (reduction #10) measures $\hat{\alpha}_c = 2.00/2.34/2.72$ at $K_{\text{eff}} \approx 16/32/64$ (alpha-vs-Gauss gap ≈ 0.7 at $K_{\text{eff}} \approx 32$: 2.34

vs gauss 3.04). The table now carries the v14 values; the v13 numbers above are kept here for provenance.

Gabor matched- K_{eff} contrast (provenance, not superseded). Table 1 reports a cross-kernel Gabor-vs-Gaussian gap of +0.31 to +0.57 at matched K_{eff} , reaching $\alpha_c \approx 3.89$ at $K_{\text{eff}} = 64$, from the v13 cross-kernel run. This is a comparison *between different kernels* (Gabor vs Gaussian) and stands as #7’s cross-kernel result. It should not be conflated with the *within-Gabor* radially control of reduction #9, which dials low-pass→band-pass via the spectral kurtosis κ_S at fixed kernel family and matched K_{eff} : that within-Gabor contrast is much smaller, $\approx +0.10$ at $K_{\text{eff}} = 32$ (band-pass higher at $K_{\text{eff}} = 64$), and was only recovered after fixing a one-line `omega0=` argv bug in #9’s first matched- K_{eff} Gabor campaign – the bug had ignored the carrier and railed every cell to $K_{\text{eff}} \approx 499$. The cross-kernel +0.3–0.6 and the within-Gabor +0.1 are not in conflict: the former measures a change of kernel *shape*, the latter a within-family *radiality knob* at fixed shape.

References

- [1] Simeon M. Berman. Excursions above high levels for stationary Gaussian processes. *Pacific Journal of Mathematics*, 36(1):63–79, 1971.
- [2] Thomas M. Cover. Geometrical and statistical properties of systems of linear inequalities with applications in pattern recognition. *IEEE Transactions on Electronic Computers*, EC-14:326–334, 1965.
- [3] Elizabeth Gardner. The space of interactions in neural network models. *Journal of Physics A: Mathematical and General*, 21:257–270, 1988.
- [4] Elizabeth Gardner and Bernard Derrida. Optimal storage properties of neural network models. *Journal of Physics A: Mathematical and General*, 21:271–284, 1988.
- [5] Robert Gütig and Haim Sompolinsky. The tempotron: A neuron that learns spike timing-based decisions. *Nature Neuroscience*, 9(3):420–428, 2006. doi: 10.1038/nn1643.
- [6] M. R. Leadbetter, Georg Lindgren, and Holger Rootzén. *Extremes and Related Properties of Random Sequences and Processes*. Springer Series in Statistics. Springer, 1983. doi: 10.1007/978-1-4612-5449-2.